

Question	Marking Guidance	Mark	Comments
04.1	Genome 1. (The) complete set of genes in a cell/organism OR (All) the DNA in a cell/organism; Proteome 2. Range of proteins that a cell/organism can produce OR Range of proteins the genome/DNA can code for;	2 (2 x AO1)	1. Accept (all) the genes/alleles/genetic material/genetic code in a cell/organism 1. Accept the total number of DNA bases in a cell/organism 1. Reject all the DNA/genes within a species/population 2. Do not accept number of proteins unqualified 2. Ignore range of proteins that a species/population can produce

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04.2	1. (The) genetic/DNA code is universal OR The same triplets/codons code for the same amino acids (in all species); 2. (The mechanism of) transcription is universal; 3. (The mechanism of) translation is universal;	2 max (2 x AO1)	1. Do not accept 'DNA is universal' unqualified 1. Reject the genetic code is degenerate 1. Ignore anything after 'genetic/DNA code is universal' unless incorrect 2. and 3. Accept descriptions of universal, eg transcription/translation are the same in humans and bacteria 2. and 3. If neither is stated, accept '(the mechanism of) protein synthesis is universal' for 1 mark 3. Accept bacteria have ribosomes, and so could translate (human mRNA)

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04.3	Cannot splice (pre-mRNA), so cannot remove introns OR Do not have Golgi (apparatus), so cannot process/modify (proteins) OR Do not have transcriptional factors (required), so cannot carry out transcription/produce mRNA;	1 (AO2)	Accept do not have spliceosomes/spliceosome for cannot splice Accept 'rough endoplasmic reticulum' for 'Golgi' Accept (human protein) is too complex and bacteria do not have Golgi (apparatus)

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04.4	1. (Region M) promoter; 2. (Region N) terminator;	2 (2 x AO2)	Accept phonetic spellings

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04.5	Shows that the (antithrombin) gene has been taken up (by cells/embryos/goats) OR Shows transgenic/transformed goat cells/goat embryos/goats OR Allows detection of genetically modified cells/organisms/mammals/goats;	1 (AO1)	Accept 'GM' for 'genetically modified'

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04.6	1. Milk/protein/antithrombin is easy to extract from a goat OR Extracting milk/protein/antithrombin from a goat does it no harm; 2. If (antithrombin was produced) in their blood, could prevent/affect clotting OR (Antithrombin) could damage other cells;	2 (2 x AO3)	